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# Toroidal Byzantine Chess

**4-ring torus (4×16 circular board with  
radial wrap)**

BOARD

**2**

PLAYERS

# TOROIDAL BYZANTINE CHESS

Toroidal Byzantine Chess combines the historical **Byzantine Chess** board (four concentric rings of 16 squares each, 64 squares total) with **toroidal topology**: the innermost ring connects to the outermost ring radially. This creates a fully closed surface with no edges – a donut-shaped (torus) board in ring form.

## BYZANTINE CHESS FOUNDATION

Byzantine Chess is a medieval circular chess variant played on a 4×16 ring board using Shatranj-era pieces:

| PIECE    | SHATRANJ NAME | MOVEMENT                              |
|----------|---------------|---------------------------------------|
| King     | Shah          | 1 step in any direction               |
| General  | Fers          | 1 step diagonally                     |
| Elephant | Fil           | Leaps exactly 2 diagonally            |
| Rook     | Rukh          | Slides orthogonally                   |
| Knight   | Faras         | Leaps as in modern chess              |
| Pawn     | Baidak        | Moves 1 step radially toward opponent |

In standard Byzantine Chess, pieces travel around the rings continuously: ring position 16 connects to position 1 on the same ring. This gives the game a circular character, with no left/right edges.

## TOROIDAL ADDITION

In Toroidal Byzantine Chess, the **radial edges** also wrap:

- A piece sliding inward from ring 4 (the innermost ring) emerges from ring 1 (the outermost ring) at the same angular position
- A piece sliding outward from ring 1 emerges from ring 4 at the same angular position
- Diagonal movement that crosses the radial boundary does so with the corresponding offset

This means there are truly no edges on the board – every ring is adjacent to two rings on both radial sides (ring 1 is adjacent to rings 2 and 4; ring 4 is adjacent to rings 3 and 1).

## CONSEQUENCES FOR PIECES

**Rooks:** can slide continuously inward/outward, wrapping from ring 4 to ring 1 and back. A Rook can reach any position on the same angular position.

**Bishops (Elephants):** the 2-step diagonal leap behaves normally within the ring structure; at the radial seam, the leap continues to the appropriate ring.

**King:** has up to 8 neighbours in all positions (no corner or edge restrictions).

## PAWNS

Pawns move radially “toward the opponent” in standard Byzantine Chess. In the toroidal version, the radial wrap means Pawns traveling inward past ring 4 emerge from ring 1 – effectively completing a full radial circuit rather than being blocked at the inner boundary.

## WIN CONDITION

Checkmate the opponent's Shah (King). Shatranj rules may also apply: stalemate = win; bare King = win (unless opponent can bare King on the next move, in which case it is a draw).

## ATTRIBUTION

Toroidal Byzantine Chess applies toroidal topology to the historical Byzantine Chess circular board. Rules derived from Byzantine Chess ([chessvariants.com/historic.dir/byzantine.html](http://chessvariants.com/historic.dir/byzantine.html)) and toroidal topology principles.